```
# Import required libraries

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import accuracy_score


# 1. Load dataset

df = pd.read_csv('Breast Cancer Dataset.csv')


# 2. Check column names (important for safety)

print(df.columns)


# 3. Preprocessing

# Drop unnecessary columns (id and unnamed if present)

df = df.drop(['id'], axis=1)


# If there is an extra unnamed column, drop it

if 'Unnamed: 32' in df.columns:

    df = df.drop(['Unnamed: 32'], axis=1)


# Separate features and target

X = df.drop(['diagnosis'], axis=1)

y = df['diagnosis']


# Convert labels: M → 0, B → 1

y = y.map({'M': 0, 'B': 1})


# 4. Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42
```

```
)

# 5. Train model
clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)

# 6. Prediction and accuracy
y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy: {:.2f}%".format(accuracy * 100))

# 7. Classify a new sample
new_sample = X_test.iloc[[0]]
prediction = clf.predict(new_sample)

# Convert prediction to readable form
prediction_class = "Malignant" if prediction[0] == 0 else "Benign"
print("New Sample Classification Result:", prediction_class)

# 8. Visualize decision tree
plt.figure(figsize=(20, 10))
plot_tree(
    clf,
    feature_names=X.columns,
    class_names=['Malignant', 'Benign'],
    filled=True,
    rounded=True,
    fontsize=10
)
plt.title("Decision Tree - Breast Cancer Dataset")
plt.show()
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

# 1. Load the dataset from your CSV file
# Ensure 'Breast Cancer Dataset.csv' is in the same folder as your notebook
df = pd.read_csv('Breast Cancer Dataset.csv')

# 2. Preprocessing
# Drop the 'id' column as it is not a predictive feature
# Separate features (X) and target (y)
X = df.drop(['id', 'diagnosis'], axis=1)
y = df['diagnosis']

# Convert categorical labels 'M' and 'B' to numbers
# M (Malignant) -> 1, B (Benign) -> 0
y = y.map({'M': 1, 'B': 0})

# 3. Split the data (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 4. Train the Decision Tree Classifier
# max_depth=3 keeps the visualization clear and readable
clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)

# 5. Make predictions and check accuracy
y_pred = clf.predict(X_test)
```

```python
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")

# 6. Classify a new sample (as requested in the manual)
# We pick the first sample from our test set as an example
# Using double brackets [[0]] passes it as a DataFrame to avoid warnings
new_sample = X_test.iloc[[0]]
prediction = clf.predict(new_sample)

# Map the numeric prediction back to a readable string
prediction_class = "Malignant" if prediction[0] == 1 else "Benign"
print(f"New Sample Classification Result: {prediction_class}")

# 7. Visualize the Decision Tree
plt.figure(figsize=(20, 10))
plot_tree(
    clf,
    feature_names=X.columns,
    class_names=['Benign', 'Malignant'],
    filled=True,
    rounded=True,
    fontsize=12
)
plt.title("Decision Tree Visualization (Breast Cancer Data)")
plt.show()
```